Information Web Retrieval and Analytics

Lecture 1: Introduction to Information Retrieval (IR)

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SEARCH
INFORMATION
GOOGLE
RETRIEVAL
USEFUL
DATABASE
DATA
DOCUMENTS
CORPUS
QUERY
DIFFICULT
MATHEMATICAL
CONFUSING
RANK
INDEX
WEB
CRAWL
DOCUMENT
LARGE
IMPORTANT
HOW
COOL
AMAZING
COMPLEX
FOR
THE
RETRIEVE
METADATA
SCIENCE
PAGE
SLOW
EFFICIENT
TEXT
LIST
FAST
FUN
SEARCHING
CRAWLER
DISTRIBUTED
GATHER
FREQUENCY
PATTERSON
MULTI
INVERTED
PROGRESSED
WEIGHTING
ONE
QUERIES
WORD
BUSINESS
QUEUING
MONEY
ENGINES
MICROSOFT
PURPLE
REUTERS
KNOWING
SORT
EVALUATE
STEMMING
COLORFUL
AGGREGATE
STATISTICS
INTRACTABLE
STUDIED
DIGGING
INTERESTING
GROWING
DISCOVERING
COSINE
ENTHRALLING
MAGICAL
TECHNOLOGICAL
SPECIAL
RELATIONAL
BIG
UNSTRUCTURED
STOP
MODIFY
TECHNICAL
CONSUMING
CHALLENGE
NEED
MATH
PARSING
DOCS
OF
AND
GET
HADOOP
ORGANIZATION
INFO
TO
IN
MAP
WAY
SCORES
EXPENSIVE
WWW
HARD
TIME
TRUST
INDEXING
ALGORITHMIC
SAD
ABOUT
NOT
ALL
VAST
TRICKY
OVERLOAD
CRAWLERS
FUTURE
COMPLICATED
STUFF
THINGS QUIRKY
SCHOLARLY
EVOLVING
SEXY
FRUSTRATING
SORTING
EXCITING
SEARCHES
EFFICIENTLY
WIKIPEDIA
RETRIEVING
MOVE
CRAWLING
KEY
DIFFERENT
VARIED
DISTRIBUTEDLY
THROUGH
LINEAR
POSTING
HAPPY
RELEVANT
POLITE
SCORE
SPEEDY
BRIGHT
NEEDED

Assessment

- Quiz and Class Activity 10%
- Assignment 10%
- Attendant 10%
- Midterm (MCQs) 30%
- Final Exam 40%



Today's agenda

- What is information retrieval?
  - Definition, documents, IR vs databases, dimensions
- IR models
  - Boolean, vector space, probabilistic
- Search engine architecture
  - Effectiveness and efficiency; indexing process vs query process
- The indexing process
  - Text acquisition, text transformation, index creation
- The query process
  - User interaction, ranking, evaluation
- Summary and Q&A

Introduction

- Information retrieval (IR) is a process that facilitates the effective and efficient retrieval of relevant information from large collections of **unstructured** or semi-structured data.

- IR systems assist in searching for, locating, and presenting information that matches a user's search query or information need.



Why is information retrieval important?

- The vast amount of data generated requires IR systems to make information accessible.
- IR is essential across industries: academia, e-commerce, healthcare, defense.
- IR acts as a human–machine interface for decision-making and knowledge discovery.
- Benefits of IR
    - **Efficient information access**: saves time and effort
    - **Knowledge discovery**: helps users find trends and patterns
    - **Personalization**: adapts results to user preferences

Main components of an IR system

- **Document collection**: the dataset from which information is retrieved
- **Indexing component**: processes documents and maps terms to documents in an optimized structure
- **Query processor**: analyzes user queries for matching against the index
- **Ranking algorithm**: assigns relevance scores to documents
- **User interface**: allows users to submit queries and view results

What is a document?



Examples

web pages, email, books, news stories, scholarly papers, text messages, Word, PowerPoint, PDF, forum postings, etc.



Common properties

Significant text content
Some structure (e.g., title, author, date for papers; subject, sender for email)

Documents vs database records

Database record

- Well-defined fields (attributes)
- e.g., bank records: account number, balance, name, address, date of birth
- Easy to compare fields with a query: exact match

```
SELECT balance FROM accounts
WHERE account_no = 10423;
```

Document (text)

- Free text with no fixed meaning per field
- Same idea, different words: *fees* vs *charges*, *increase* vs *rise*
- Matching text is much harder: a matter of degree

```
query: bank fees increase
```

IR vs databases

	Database search	Information retrieval
Data	Structured records (tables)	Unstructured text (documents)
Query	SQL: formal and precise	A few words: vague
Matching	Exact: a row matches or not	Partial: degrees of relevance
Output	A set of rows	A ranked list
Structure	Tables, B-tree indexes	Inverted index
Example	account_no = 10423	cheap flight Phnom Penh Paris

After Manning, Raghavan & Schütze (2008), Table 10.1

Dimensions of IR

Content	Applications	Tasks
Text	Web search	Ad hoc search
Images	Vertical search	Filtering
Video	Enterprise search	Classification
Scanned docs	Desktop search	Question answering
Audio	Forum search	
Music	P2P search	
	Literature search	

Croft, Metzler & Strohman (2010), ch. 1

Types of information retrieval models

- **Boolean model**: uses Boolean logic (AND, OR, NOT) for exact matching; no ranking
- **Vector space model**: represents documents and queries as vectors, using cosine similarity to rank results by relevance
- **Probabilistic model**: estimates the probability that a document is relevant; leads to BM25

How does a modern IR system work? Vector search

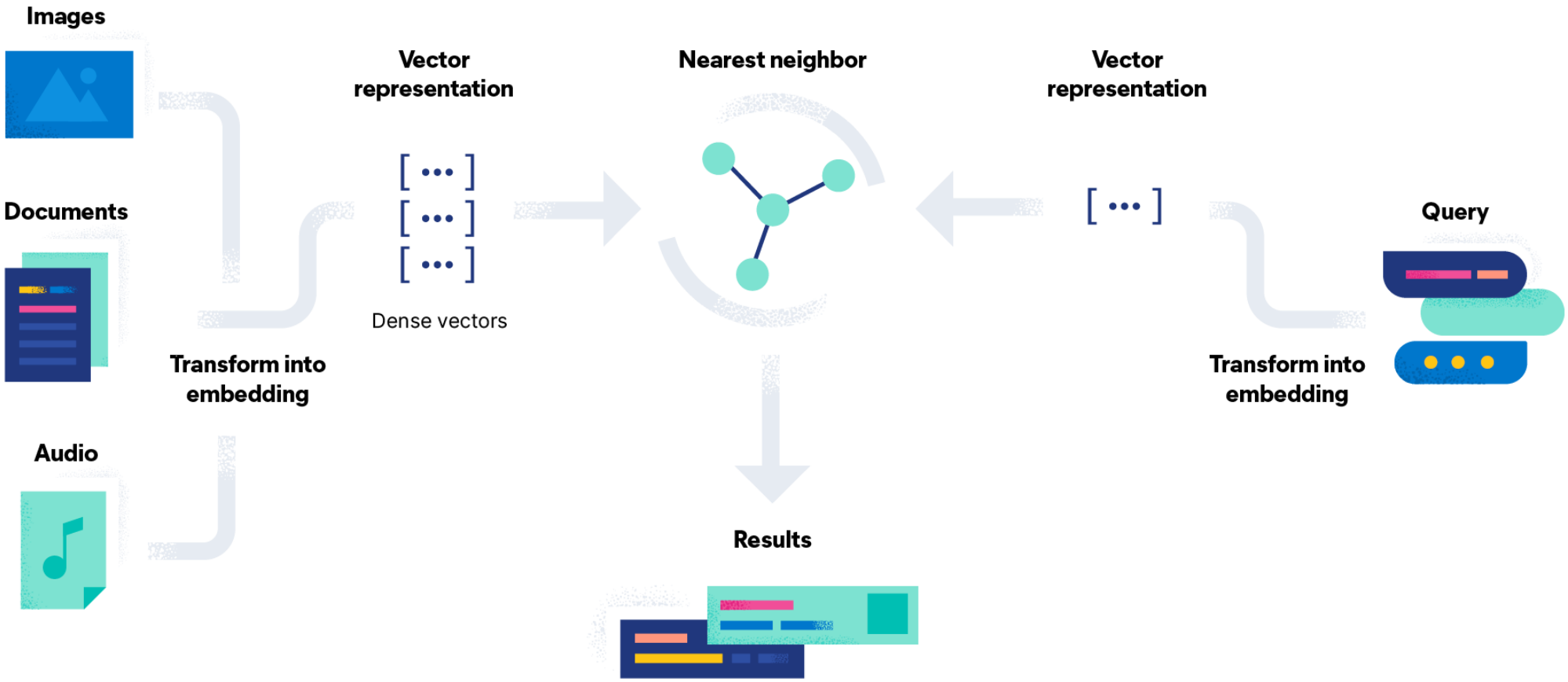


Figure: Elastic, “What is vector search?” (elastic.co/what-is/vector-search);

IR and search engines

Information Retrieval	Search Engines
Relevance: effective ranking	Performance: efficient search and indexing
Evaluation: testing and measuring	Incorporating new data: coverage and freshness
Information needs: user interaction	Scalability: growing with data and users
	Adaptability: tuning for applications
	Specific problems: spam

Croft, Metzler & Strohman (2010), ch. 1

Search engine architecture

- A software architecture consists of software components, the interfaces provided by those components, and the relationships between them
- Describes a system at a particular level of abstraction
- The architecture of a search engine is determined by two requirements
  - **Effectiveness**: quality of results (precision, recall)
  - **Efficiency**: response time and throughput (latency, queries per second)
- Designing a search engine is a constant trade-off between the two

Indexing process

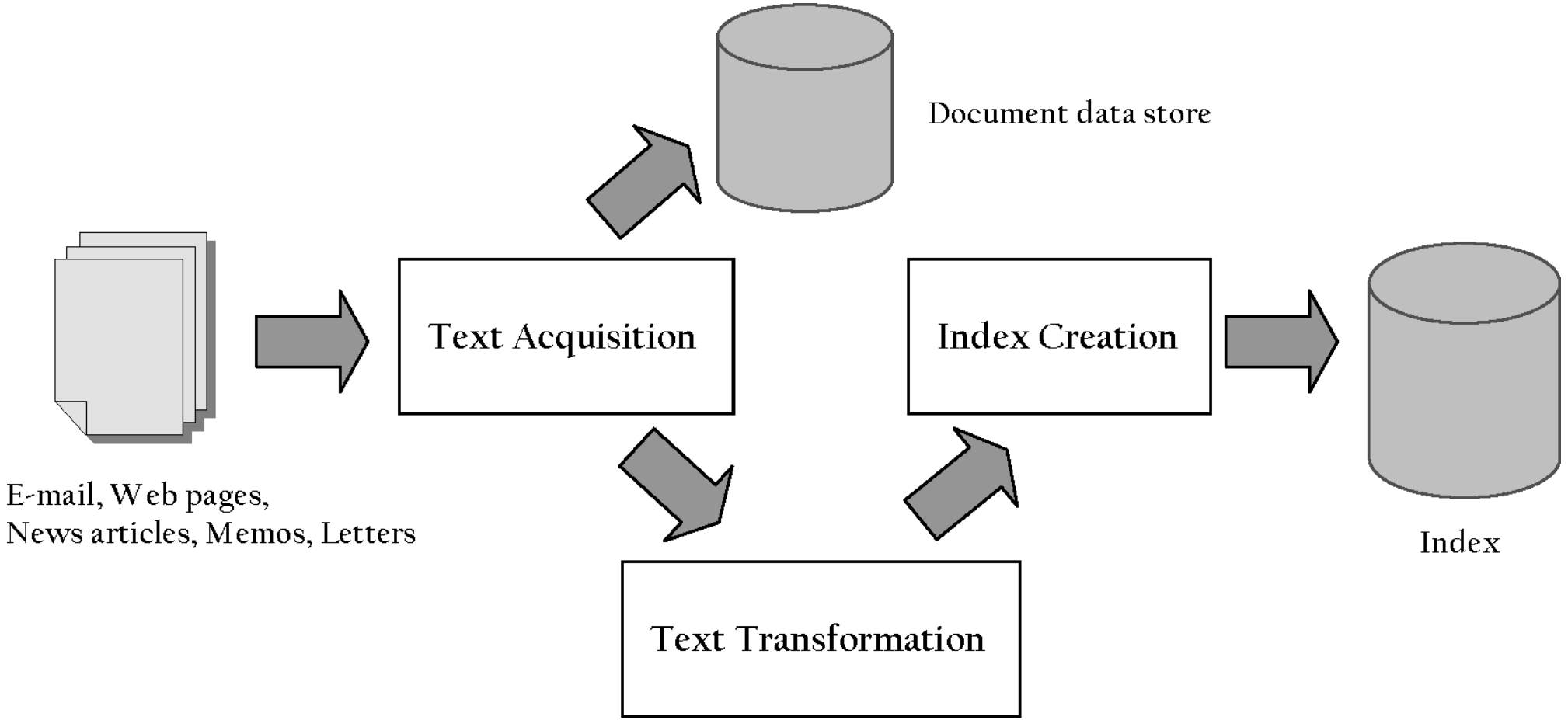


Figure 2.1 in Croft, Metzler & Strohman (2010), *Search Engines: Information Retrieval in Practice*

Indexing process (cont.)

- **Text acquisition**
  - identifies and stores documents for indexing
- **Text transformation**
  - transforms documents into index terms or features
- **Index creation**
  - takes index terms and creates data structures (indexes) to support fast searching

Query process

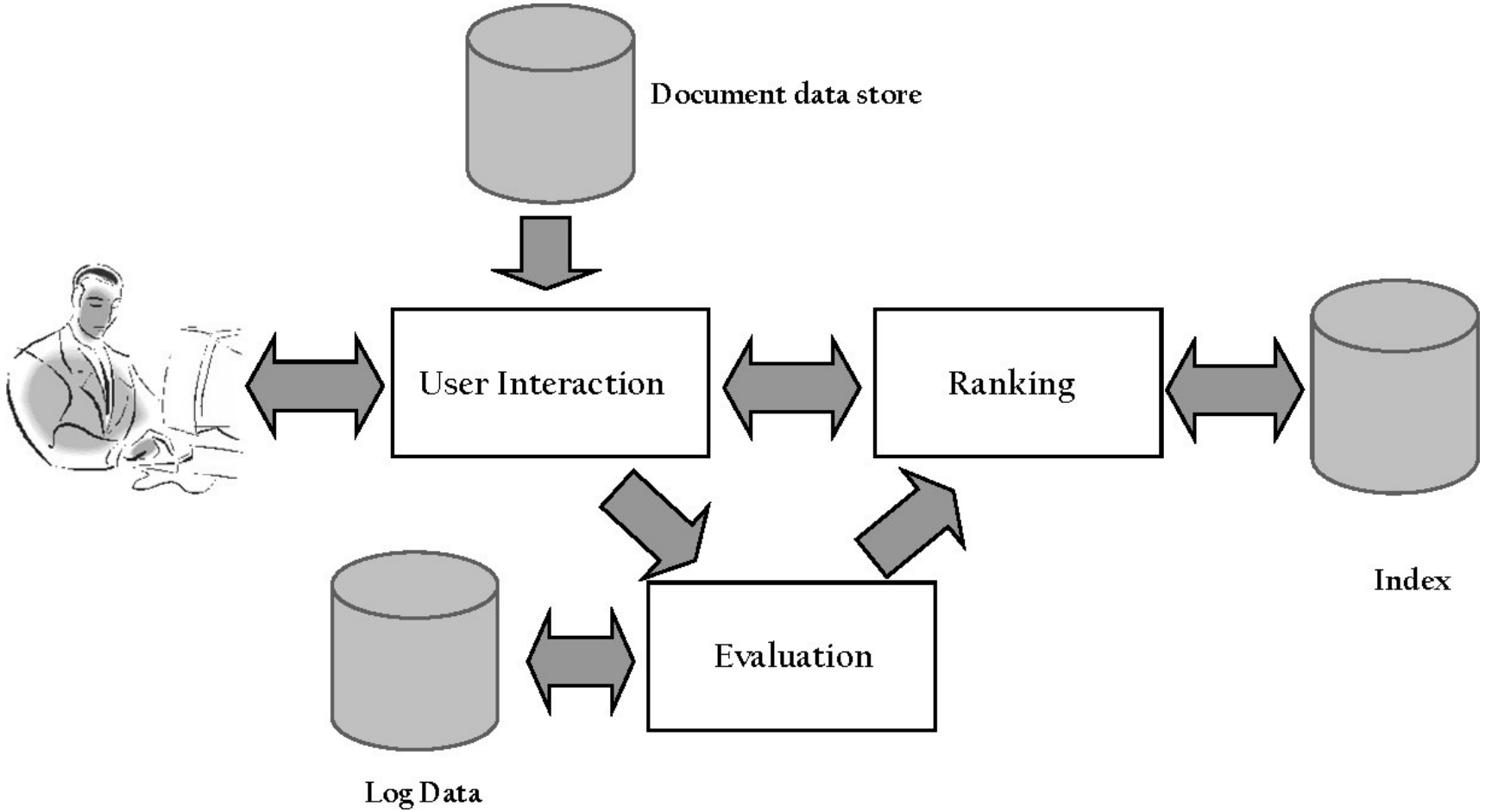


Figure 2.2 in Croft, Metzler & Strohman (2010), *Search Engines: Information Retrieval in Practice*

Query process (cont.)

- **User interaction**
  - supports creation and refinement of the query, and display of results
- **Ranking**
  - uses the query and indexes to generate a ranked list of documents
- **Evaluation**
  - monitors and measures effectiveness and efficiency
  - logging happens online; ranking and performance analysis run offline on logs and test collections

Text acquisition

- **Crawler**: identifies and acquires documents for the search engine
  - Many types: web, enterprise, desktop
  - Web crawlers follow links to find documents
    - Must efficiently find huge numbers of web pages (coverage) and keep them up to date (freshness)
    - Single-site crawlers for site search
    - Topical or focused crawlers for vertical search
  - Document crawlers for enterprise and desktop search: follow links and scan directories

Text acquisition (cont.)

- **Feeds**
  - Real-time streams of documents, e.g., web feeds for news, blogs, video, radio, TV
  - RSS is a common standard
    - An RSS “reader” can provide new XML documents to the search engine

- **Conversion**
  - Convert a variety of documents into a consistent text plus metadata format
    - e.g., HTML, XML, Word, PDF, etc. → XML
  - Convert text encoding for different languages, using a Unicode standard like UTF-8

Text acquisition (cont.)

- **Document data store**
  - Stores text, metadata, and other related content for documents
    - Metadata is information about a document, such as type and creation date
    - Other content includes links and anchor text
  - Provides fast access to document contents for search engine components
    - e.g., result list generation (titles, snippets)
  - Could use a relational database system
    - More typically, a simpler, more efficient storage system is used due to huge numbers of documents

Text transformation

- **Parser**: processes the sequence of text tokens to recognize structural elements
  - e.g., titles, links, headings; uses markup syntax such as <h2>Overview</h2>
- **Tokenizer**: recognizes “words” in the text
  - must handle capitalization, punctuation, non-alpha characters, separators

Step	Output
Raw	The computers are computing the results.
Tokenize	the · computers · are · computing · the · results
Stop	computers · computing · results
Stem	comput · comput · result

Text transformation (cont.)

- **Link analysis**: makes use of links and anchor text in web pages
  - Identifies popularity and community information, e.g., PageRank
  - Anchor text can significantly enhance the representation of pages pointed to by links
  - Significant impact on web search; less importance in other applications
- **Information extraction**: identify classes of index terms important for some applications
  - e.g., named entity recognizers identify people, locations, companies, dates
- **Classifier**: identifies class-related metadata for documents
  - assigns labels: topics, reading levels, sentiment, spam

Index creation

- **Document statistics**
  - Gathers counts and positions of words and other features
  - Used in the ranking algorithm
- **Weighting**
  - Computes weights for index terms, used in the ranking algorithm
  - e.g., tf·idf: term frequency in the document × inverse document frequency in the collection
- **Inversion**
  - Core of indexing: converts document → terms into term → documents (the inverted index)
    - Difficult for very large numbers of documents
  - Format designed for fast query processing
    - Must also handle updates
    - Compression used for efficiency

Index creation (cont.)

- **Index distribution**
  - Distributes indexes across multiple computers and multiple sites
  - Essential for fast query processing with large numbers of documents
  - Many variations such as:
    - **Document distribution**: each machine indexes a subset of the documents
    - **Term distribution**: each machine holds postings for a subset of the terms
    - **Replication**: copies of the index serve more queries and survive failures
  - P2P and distributed IR involve search across multiple sites

User interaction

- **Query input**
  - Provides the interface and parser for the query language
  - Most web queries are very simple; other applications may use forms
  - Query languages describe more complex queries, e.g., Boolean: “information retrieval” AND (course OR lecture)
- **Query transformation**
  - Improves the initial query, both before and after the initial search
  - Spell checking and query suggestion: retreival → retrieval
  - Query expansion and relevance feedback add terms: car → car OR automobile
- **Results output**
  - Constructs the display of ranked documents for a query
  - Generates snippets to show how queries match documents
  - Highlights important words and passages
  - Retrieves appropriate advertising in many applications
  - May provide clustering and other visualization tools

Ranking

- **Scoring**
  - Calculates scores for documents using a ranking algorithm: the core component of a search engine
  - Basic form of the score is $q_i d_i$
    - q_i and d_i are the query and document term weights for term i
  - Many variations of ranking algorithms and retrieval models
- **Performance optimization**
  - Designing ranking algorithms for efficient processing
  - Term-at-a-time vs document-at-a-time processing
  - Safe vs unsafe optimizations
- **Distribution**
  - Processing queries in a distributed environment
  - Query broker distributes queries and assembles results
  - Caching is a form of distributed searching

Evaluation

- **Logging**
  - Logging user queries and interaction is crucial for improving search effectiveness and efficiency
  - Query logs and clickthrough data are used for query suggestion, spell checking, query caching, ranking, advertising search, and other components
- **Ranking analysis**
  - Measuring and tuning ranking effectiveness
- **Performance analysis**
  - Measuring and tuning system efficiency

Summary

- IR retrieves **relevant** information from large collections of unstructured data, effectively and efficiently
- Documents are free text, so IR ranks by degree of relevance instead of matching exactly like a database
- Models range from Boolean sets to vectors, and probabilities
- A search engine = an **indexing process** (acquire, transform, index) + a **query process** (interact, rank, evaluate)

References

- Croft, W. B., Metzler, D., & Strohman, T. (2010). *Search Engines: Information Retrieval in Practice*. Pearson.
- Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.
- Elastic. What is vector search? elastic.co/what-is/vector-search (vector search figure).



Thank you!